

## Data Collection and Preprocessing Phase

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|---------------|------------------------------------------|
| Date          | 21 September 2026                        |
| Team ID       | MCA                                      |
| Project Title | Smartphone Addiction Analytics Dashboard |
| Maximum Marks | 10 Marks                                 |

## Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

| Section       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| Data Overview | Smartphone_Usage.csv contains 7,500 records and 16 columns describing young adult users (age 18–35). Key fields: demographics (age, gender), usage behaviour (daily/weekend screen time, social media, gaming, work/study hours, notifications, app opens), lifestyle (sleep hours, stress level, academic/work impact) and target variables (addiction_level, addicted_label). Source is a structured CSV suitable for descriptive and diagnostic analytics. |
| Data Cleaning | No duplicate rows found (0 duplicates). Only addiction_level has missing values (819 NaNs ≈ 10.9%). Plan: (1) keep the rows for completeness of other metrics, (2) create a binary flag is_addiction_missing, (3) for modelling/dashboard                                                                                                                                                                                                                     |

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|                              | filters treat missing as a separate category 'Unknown'. All other columns are complete. No obvious data-entry errors detected in numeric ranges.                                                                                                                                                                                                                                                                                                  |
| Data Transformation          | In Power Query / pandas: (1) filter out any future invalid ranges if needed, (2) create calculated columns – total_entertainment_hours = social_media_hours + gaming_hours, screen_to_sleep_ratio = daily_screen_time_hours / sleep_hours, high_notification_flag = notifications_per_day > 150. (3) Pivot / group-by summaries for age-group and gender will be created for dashboard visuals. Sorting by addiction severity for priority views. |
| Data Type Conversion         | age, notifications_per_day, app_opens_per_day, addicted_label already integer. All hour columns already float. Categorical columns (gender, stress_level, academic_work_impact, addiction_level) converted to category / ordered category where appropriate (e.g. addiction_level ordered as None < Mild < Moderate < Severe). transaction_id and user_id kept as string identifiers.                                                             |
| Column Splitting and Merging | No splitting required – all columns are atomic. Optional merge: create a derived dimension table for user demographics (user_id, age, gender) if moving to a star-schema model later. For current single-table analysis no merge is necessary.                                                                                                                                                                                                    |
| Data Modeling                | Single fact table is sufficient for the dashboard. If expanded later: User dimension (user_id, age, gender) linked to Fact_Usage on user_id. Measures planned: % Severe addiction, Avg daily screen time, Avg notifications, High-risk user count, Correlation coefficients between key numeric predictors and addicted_label.                                                                                                                    |

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| Save Processed Data | Cleaned dataset saved as Smartphone_Usage_Cleaned.csv (and optionally .parquet). A second version with imputed / flagged addiction_level and calculated columns is stored as Smartphone_Usage_Processed.csv for dashboard consumption. Version control via Git; original raw file retained unchanged. |
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